

1. Document Information

Field	Value
Project Name	RMG Manufacturing & Business Performance Dashboard
Project Type	Data Analytics & Business Intelligence
Industry	RMG Sector
Role	Data Analyst
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Tools	PostgreSQL, Power BI, DAX
Date	20-Aug-2026
Status	Completed

2. Business Requirement Overview

The business needs a clear and centralized way to monitor order, shipment, production, factory, and quality performance.

The available data contains information across multiple business processes, but reviewing these areas separately makes it difficult to identify performance gaps and compare business and operational performance.

The proposed solution is an interactive Power BI dashboard that brings these areas together and provides a consistent view of key business metrics and performance indicators.

3. Business Goals

The dashboard should help users to:

- Monitor overall business and order performance.
- Understand buyer and order value performance.
- Evaluate shipment performance and freight costs.
- Compare production performance across factories and production lines.
- Monitor production efficiency and achievement.
- Identify quality and process-related performance issues.
- Explore trends and performance differences using interactive filters.

4. Functional Requirements

FR-01: Executive Performance Monitoring

The dashboard must provide a high-level view of overall business performance, including:

- Total Order Value
- Total Orders
- Production Achievement %
- On-Time Shipment %

Users should be able to quickly understand the overall business situation from the Executive Overview page.

FR-02: Order & Buyer Analysis

The dashboard should allow users to analyze:

- Order value trends
- Order quantity by buyer
- Top buyers by order value
- Average Order Value

Users should be able to compare buyer performance and identify major contributors to order value.

FR-03: Shipment Performance Analysis

The dashboard should provide visibility into:

- On-Time Shipment %
- Planned vs Actual Shipments
- Freight Cost by Factory
- Total Freight Cost

This should help users understand shipment performance and identify differences in freight costs across factories.

FR-04: Production Performance Analysis

The dashboard should allow users to evaluate:

- Actual Production

- Production Achievement %
- Production Efficiency %
- Production performance by factory
- Production output by production line
- Lost Time %

The analysis should support comparison of production performance across factories and production lines.

FR-05: Quality & Process Analysis

The dashboard should provide visibility into key quality and process metrics, including:

- Finishing Reject %
- Finishing Reject Quantity by Factory
- Dyeing Batch Loss %
- Dyeing Reprocess %
- Dyeing Reprocess Rate by Factory
- Finishing Packing Rate %
- Finished vs Packed Quantity
- Monthly Reject Quantity

This should help users identify quality-related performance gaps and process-level issues.

5. KPI Requirements

The dashboard must use consistent KPI definitions across the relevant pages.

Executive KPIs

- Total Order Value
- Total Orders
- Production Achievement %
- On-Time Shipment %

Order, Buyer & Shipment KPIs

- Average Order Value
- Total Freight Cost
- On-Time Shipment %

Production & Factory KPIs

- Actual Production
- Production Efficiency %

- Production Achievement %
- Lost Time %

Quality & Process KPIs

- Finishing Reject %
- Dyeing Batch Loss %
- Dyeing Reprocess %
- Finishing Packing Rate %

6. Dashboard Requirements

The dashboard should be organized into four pages.

Page 1 — Executive Overview

The page should provide an overall business snapshot through:

- Order value trend
- Shipment status
- Monthly production target vs actual
- Production by factory

Page 2 — Order, Buyer & Shipment

The page should focus on:

- Buyer performance
- Order value and quantity
- Freight cost by factory
- Planned vs actual shipments

Page 3 — Production & Factory Performance

The page should focus on:

- Production achievement by factory
- Production output by line
- Production efficiency by factory
- Production performance by factory

Page 4 — Quality & Process Performance

The page should focus on:

- Finishing rejects
- Finished vs packed quantities
- Dyeing reprocessing
- Monthly reject trends

7. Filtering & Interaction Requirements

The dashboard should provide interactive filters that allow users to explore performance from different perspectives.

Required Filters

- Year
- Factory

- Buyer — where relevant to the page
- Filters should update the relevant KPIs and visualizations dynamically.
The dashboard should maintain consistent filtering behavior across related visuals on each page.

8. Visualization Requirements

The dashboard should use appropriate visualizations based on the type of analysis.

- KPI Cards for high-level performance metrics
- Line Charts for monthly trends
- Bar Charts for factory and buyer comparisons
- Clustered Column Charts for planned vs actual comparisons
- Donut Chart for shipment status

Visuals should be simple, readable, and focused on business insights rather than unnecessary decoration.

9. Data & Calculation Requirements

The solution should:

- Use clean and structured data for reporting.
- Maintain appropriate relationships between fact and dimension tables.
- Use DAX measures for KPI calculations.
- Apply consistent KPI definitions across the dashboard.
- Ensure numerical and percentage values are displayed with appropriate formatting.
- Allow dashboard metrics to respond correctly to applied filters.

10. Business Acceptance Criteria

The project will be considered successful when:

- All four dashboard pages are functional.
- Required KPIs are correctly calculated and displayed.
- Dashboard filters work as expected.
- Users can compare buyers, factories, production lines, and quality performance.
- Monthly trends can be analyzed easily.
- Planned and actual performance can be compared where applicable.
- The dashboard provides a clear and consistent view of business and operational performance.

11. Expected Business Value

The dashboard is expected to provide a consolidated view of business and operational performance, making it easier to identify performance differences, monitor trends, and focus on areas that may require further investigation.